Chromatin Accessibility Changes Across Developmental Time Points in Human Corticogenesis

1. Introduction

Corticogenesis, the development of the human cerebral cortex, is a complex, tightly regulated process beginning around post-conception weeks (PCW) 4–6 and extending into late gestation. During this period, diverse neural progenitor populations proliferate, differentiate, and establish the early architecture of the cortex. Importantly, many neurodevelopmental disorders originate during mid-gestation, particularly between PCW16–24, a window associated with rapid neurogenesis, progenitor activity, and dynamic transcriptional regulation.

Chromatin accessibility reflects the openness of genomic regulatory elements and reveals which regions are available for transcription factor binding and gene activation. Single-cell ATAC-seq (scATAC-seq) provides a high-resolution view of chromatin states across distinct cell types and developmental time points.

This study evaluates how chromatin accessibility shifts during mid-gestational corticogenesis. Two primary research questions/aims guided the analysis:

(1) How does global chromatin accessibility change between PCW16 and PCW24?

(2) Which regulatory regions gain or lose accessibility across these developmental transitions?

2. Dataset Overview

The analysis utilized the GSE162170 single-cell multiomic dataset. This was a subset of a comprehensive study profiling human cortical development (Trevino et al.). From this resource, scATAC-seq data were extracted, including a peak-by-cell accessibility matrix, genomic coordinates for consensus peaks, and detailed metadata. To balance computational feasibility with biological representation, 3,000 cells were randomly subsampled, ensuring coverage across all developmental ages included in the study. This subsampling preserved stage diversity while enabling efficient peak-level and cell-level analyses.

3. Methods

3.1 Aim 1: Quantifying Global Chromatin Accessibility Across Development

Low-accessibility peaks were filtered out, retaining only those present in at least 1% of all cells, which resulted in approximately 219,480 high-confidence peaks. Cell-level accessibility was then quantified by calculating total ATAC counts per cell and applying log10 normalization to reduce overdispersion. For dimensionality reduction, TF-IDF normalization and singular value decomposition (SVD) were performed, followed by UMAP to visualize global accessibility structure across time points. A Kruskal–Wallis test was used to assess statistical differences in global chromatin accessibility between developmental stages.

3.2 Aim 2: Identifying Differentially Accessible Regions and Regulatory Programs

Differential accessibility (DA) analysis was performed using logistic regression while modeling sequencing depth as a latent covariate. PCW16 was used as the baseline reference stage, and comparisons were made to PCW20, PCW21, and PCW24. Peaks were considered significantly different if they met an FDR threshold of less than 0.05 and exhibited an absolute $\log_2$ fold change greater than 0.25. Significant peaks were then annotated to their nearest genes using TxDb.Hsapiens.UCSC.hg38 genomic annotations. Lastly, motif enrichment analysis was conducted using JASPAR 2020 position frequency matrices and implemented through Signac's *FindMotifs* function to identify transcription factor motifs enriched among differentially accessible peaks.

4. Results

4.1 Global Accessibility Across Development (Aim 1)

Analyses revealed that total chromatin accessibility per cell increased progressively from PCW16 to PCW24. This general upward trend suggests that maturing cortical cells open more regulatory regions as they differentiate and adopt specialized functions. The Kruskal-Wallis result ($p = 2.2\times10^{-16}$) reinforces that

observed differences in global accessibility between stages are statistically significant and unlikely due to technical noise.

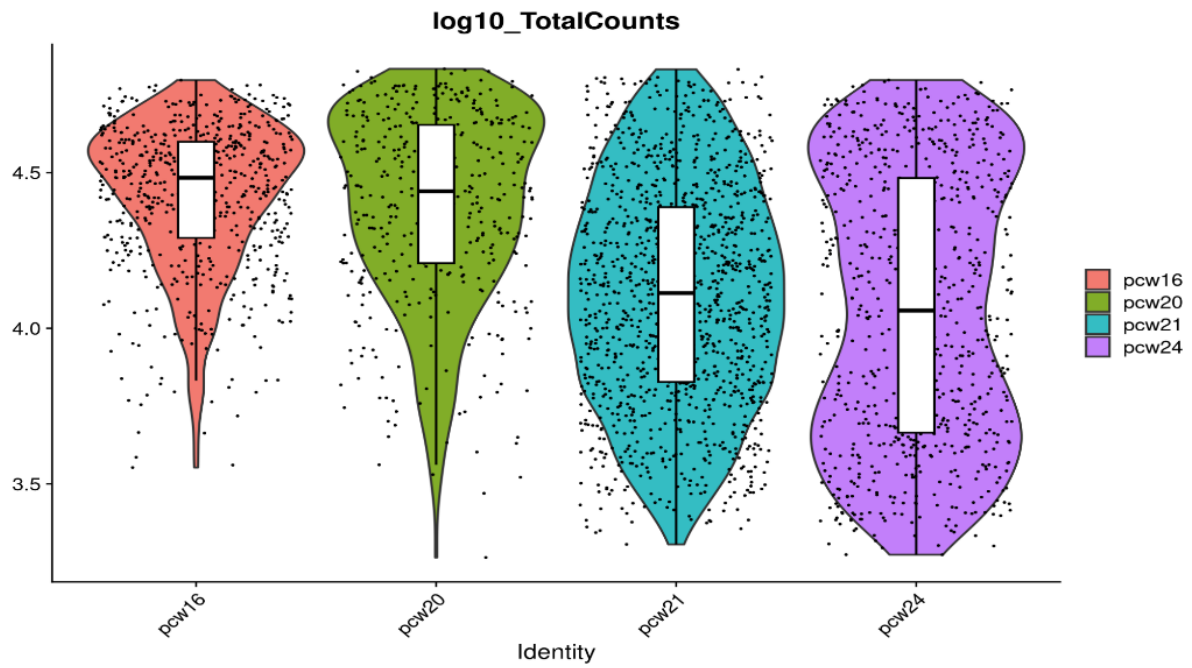

Fig 1: Violin Plot of Global Chromatin Accessibility across Developmental Stages

UMAP embeddings (Fig 2) showed moderately distinct clustering, reflecting slightly substantial shifts in overall chromatin landscape from PCW16 to other developmental stages. Datapoints for PCW-20, -21, and -24 seemed to overlap in the UMAP embeddings, suggesting that these later stages share broadly similar global accessibility profiles with only subtle stage-specific differences.

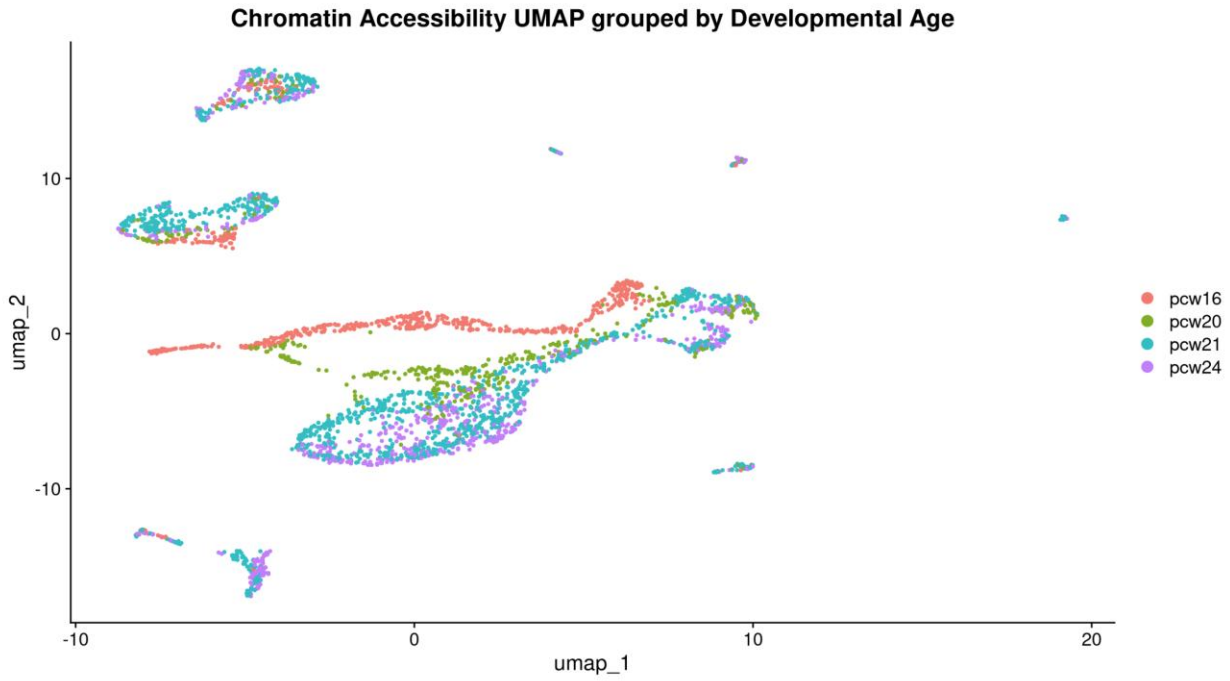

Fig 2: Age-Stratified Chromatin Accessibility Embeddings

4.2 Differential Accessibility at Key Developmental Transitions (Aim 2)

Differential accessibility (DA) analysis identified numerous peaks exhibiting significant accessibility changes relative to PCW16, with the magnitude of these changes increasing progressively across developmental time (Fig 3). PCW20 showed the fewest DA peaks, reflecting an early and relatively modest shift in chromatin accessibility consistent with its close developmental proximity to PCW16. PCW21 displayed an intermediate number of DA regions, indicative of ongoing chromatin remodeling and the emergence of stage-specific regulatory elements. By contrast, PCW24 exhibited the largest number of significant DA peaks, signaling extensive chromatin reorganization during this peak neurogenic period.

Fig 3: Comparative Volcano Plot of DA Peaks by Developmental Stage

4.3 Regulatory and Motif-Level Insights (Aim 2)

Top 5 Enriched Motifs per Developmental Stage

	Stage	Motif ID	Transcription Factor	Fold Enrichment	Adjusted P-value	# Peaks
MA1524.1	PCW20	MA1524.1	MSGN1	1.938	0	110
MA0671.1	PCW20	MA0671.1	NFIX	1.857	0	134
MA0623.2…3	PCW20	MA0623.2	NEUROG1	1.843	0	76
MA0818.1…4	PCW20	MA0818.1	BHLHE22	1.835	0	70
MA0668.1…5	PCW20	MA0668.1	NEUROD2	1.835	0	69
MA0668.1…6	PCW21	MA0668.1	NEUROD2	3.010	0	353
MA0623.2…7	PCW21	MA0623.2	NEUROG1	2.933	0	378
MA0826.1…8	PCW21	MA0826.1	OLIG1	2.884	0	366
MA0818.1…9	PCW21	MA0818.1	BHLHE22	2.879	0	344
MA0817.1	PCW21	MA0817.1	BHLHE23	2.687	0	367
MA0668.1…11	PCW24	MA0668.1	NEUROD2	2.542	0	361
MA0623.2…12	PCW24	MA0623.2	NEUROG1	2.359	0	368
MA0826.1…13	PCW24	MA0826.1	OLIG1	2.340	0	362
MA0818.1…14	PCW24	MA0818.1	BHLHE22	2.272	0	340
MA0827.1	PCW24	MA0827.1	OLIG3	2.252	0	410

Table 1: Top 5 Enriched Transcription Factor Motifs Across Developmental Stages

Annotation of DA peaks to their nearest genes revealed strong enrichment for genes associated with neuronal differentiation and specification (motifs for *NEUROG1* and *NEUROD2*, bHLH transcription

factors that promote neuronal lineage commitment and differentiation) and neurogenesis more broadly, as well as factors implicated in neural patterning and regional identity (motifs for *NFIX* and *OLIG* family TFs, known regulators of neural progenitor fate and patterning), highlighting the biological relevance of the identified accessibility changes (Velkey et al.).

5. Discussion & Limitations

This work provides evidence that mid-gestational corticogenesis is marked by broad and dynamic changes in chromatin regulation. We observed global increases in chromatin accessibility across developmental stages, suggesting enhanced regulatory activity as cells progress through differentiation. Differential accessibility analyses further revealed stage-specific regulatory changes, with PCW24 exhibiting the most extensive chromatin remodeling, consistent with this period representing a peak in neurogenic activity. Motif enrichment supported the activation of neurogenic transcription factor networks, indicating that accessibility changes reflect coordinated engagement of developmental programs rather than random fluctuations.

Despite these insights, several limitations temper the interpretation of the results. First, the subsampling of 3,000 cells reduces sensitivity for detecting rare cell populations and subtle regulatory events. Random sampling also introduces variance that may slightly influence UMAP structure and the detection of differentially accessible peaks. The developmental spacing of the samples poses an additional challenge, as the transition from PCW20 to PCW21 spans only a single week, meaning that small differences should be interpreted cautiously. Limitations in peak annotation also affect interpretation; nearest-gene assignment does not fully capture the non-linear nature of enhancer–gene interactions, particularly given the absence of chromatin-looping information. Finally, potential batch and donor effects remain a concern, and future analyses incorporating mixed-effect models or donor-level covariates would help mitigate residual confounding.

7. References

1. Trevino AE, Müller F, Andersen J, Sundaram L, Kathiria A, Shcherbina A, Farh K, Chang HY, Pașca AM, Kundaje A, Pașca SP, Greenleaf WJ. Chromatin and gene-regulatory dynamics of the developing human cerebral cortex at single-cell resolution. Cell. 2021 Sep 16;184(19):5053-5069.e23. doi: 10.1016/j.cell.2021.07.039. Epub 2021 Aug 13. PMID: 34390642.

2. Velkey JM, O'Shea KS. Expression of Neurogenin 1 in mouse embryonic stem cells directs the differentiation of neuronal precursors and identifies unique patterns of down-stream gene expression. *Dev Dyn*. 2013;242(3):230-253. doi:10.1002/dvdy.23920